RELATIVE DIMENSION AT A POINT OF THE TARGET EQUALS THE FIBRE DIMENSION FOR MAPS OF SPACES

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7123-7135 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 7137-7180, the complete author proof (42 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. The dictionary between the relative-dimension and fibre-dimension viewpoints for spaces; every one of the author's replacements is backed by a named lemma, the final equality is by definition.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Definitions of fibre dimension, dimension of a space, and dimension at a point (Properties of Spaces definition section, Morphisms of Spaces definition dimension-fibre); etale invariance of space dimension (Properties of Spaces, lemma dimension-decent-invariant-under-etale); base-change invariance of fibre relative dimension (Morphisms of Spaces, lemma dimension-fibre-after-base-change); the field-etale structure lemma (Morphisms, lemma etale-over-field).

Source reference. Original lemma 02GL: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7180–7180. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NM: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7180–7180. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04NS: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7180–7180. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04N6: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7180–7180. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 04N5: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7180–7180. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0A4H: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7180–7180. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 03H4: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7180–7180. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a finite type morphism of algebraic spaces over S. Let $y \in |Y|$. The following quantities are the same*

(1) *$d = -\infty$ if y is not in the image of $|f|$ and otherwise the minimal integer d such that f has relative dimension $\leq d$ at every $x \in |X|$ mapping to y,*
(2) *the dimension of the algebraic space $X_k = \mathrm{Spec}(k) \times_Y X$ for any morphism $\mathrm{Spec}(k) \to Y$ in the equivalence class defining y.*

Proof. To parse this one has to consult Morphisms of Spaces, Definition 04NM, Properties of Spaces, Definition 04N6, Properties of Spaces, Definition 04N5. We will show that the numbers in (1) and (2) are equal for a fixed morphism $\mathrm{Spec}(k) \to Y$. Choose an étale morphism $V \to Y$ where V is an affine scheme and a point $v \in V$ mapping to y. Since $V \times_Y \mathrm{Spec}(k) \to \mathrm{Spec}(k)$ is surjective étale (by Properties of Spaces, Lemma 03H4) we can find a finite separable extension k'/k (by Morphisms, Lemma 02GL) and a commutative diagram

$$\begin{array}{ccc} \mathrm{Spec}(k') & \longrightarrow & V \\ \downarrow & & \downarrow \\ \mathrm{Spec}(k) & \longrightarrow & Y \end{array}$$

We may replace $X \to Y$ by $V \times_Y X \to V$ and X_k by $X_{k'} = \mathrm{Spec}(k') \times_V (V \times_Y X)$ because this does not change the dimensions in question by Properties of Spaces, Lemma 0A4H and Morphisms of Spaces, Lemma 04NS. Thus we may assume that Y is an affine scheme. In this case we may assume that $k = \kappa(y)$ because the dimension of $X_{\kappa(y)}$ and X_k are the same by the aforementioned Morphisms of Spaces, Lemma 04NS and the fact that for an algebraic space Z over a field K the relative dimension of Z at a point $z \in |Z|$ is the same as $\dim_z(Z)$ by definition. Assume Y is affine and $k = \kappa(y)$. Then X is quasi-compact we can choose an affine scheme U and an surjective étale morphism $U \to X$. Then $\dim(X_k) = \dim(U_k) = \max \dim_u(U_k)$ is equal to the number given in (1) by definition. □